

Do Analyst Revisions Add Anything to Industry Momentum?

A falsification test across the 49 Fama–French industries

MFE 230GA Active Asset Management — Final Project

Group 4: Hashim Almodamagha, Aditya Aryan, Jack Duncan,

Charishma Takkallapalli, Cristhian Ruiz Cardozo

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Executive Summary

We test whether industry-level analyst EPS revisions predict next-month returns across the 49 Fama–French industries *beyond* industry momentum. The mechanism is slow diffusion of earnings news: if prices absorb analysts’ upgrades gradually, industries with net upgrades should outperform next month even after controlling for past returns. We built monthly industry revision scores from I/B/E/S (1985–2025, linked to CRSP), industry momentum from Ken French returns, blended the two with expanding IC-based weights (the HW2 rule $w \propto R^{-1}$ IC), and sized dollar-neutral long-short books with a mean-variance optimizer on a shrunk EWMA covariance (30-month half-life, as in HW1) at 5% ex-ante active risk, charging 10–30 bp one-way costs. Before testing we fixed the rule: if no strategy earns FF5+UMD alpha with $t \geq 2$ net of costs, especially after 2010, we would not implement. **The result is negative.** Momentum has a real IC (0.047, Newey–West $t = 3.96$, 1985–2025). Revisions are weak alone (IC 0.010, $t = 1.06$) and carry no information once momentum is removed (IC -0.000 , $t = -0.05$). Revision portfolios trade about 175% of capital a month and lose 4–5% a year after costs. The blend looks like alpha against FF5 (2.4% a year, $t = 2.21$), but that alpha falls to 0.2% ($t = 0.36$) once UMD is added, and to -0.0% after 2010. No robustness setting reaches $t \geq 2$. **Recommendation: do not implement the analyst-revision overlay in its current form; the blended portfolio’s performance is momentum exposure, not incremental revision alpha.** This strategy is our second project: we turned to it after our first idea, a climate-attention timing strategy built on Homework 1, failed its own holdout and robustness tests (separate research note).

1 What Did We Try?

1.1 Research path: from Climate Alpha to analyst revisions

Our first project continued the Homework 1 emissions exercise: a factor-hedged short of high-emission industries, timed by a “climate-transition attention” series. It looked good in its design window but lost money in an untouched 2022–2026 holdout, failed a frozen 1994–2009 test, and its signal turned out to be a stock-market-volatility news index rather than a climate measure. We did not force it; the evidence is in *Climate Alpha: The Hypothesis We Tested and Rejected* (folder 01_climate_alpha_pivot). One lesson carried over: every apparent industry alpha in that work was a known factor in disguise, and industry momentum in particular was almost entirely UMD. That shaped the present question, where the decisive test is whether a signal survives UMD.

Connection to the homeworks. HW1 gave us the risk machinery we reuse here: monthly value-weighted Ken French returns, a half-life-weighted covariance (30 months), volatilities and betas, and the habit of checking an AI-proposed idea against our own estimates. HW2 gave us the alpha and sizing machinery: alphas from IC and residual risk ($\alpha = \text{IC} \cdot \omega \cdot z$), blending correlated signals with $w \propto R^{-1}\text{IC}$, holdings $h_n = \alpha_n / (2\lambda\omega_n^2)$, and the critique of assuming a fixed alpha. The final strategy is not the HW1 strategy; the connection is the research path and the tools.

1.2 Thesis and falsification criterion (fixed before testing)

Earnings news diffuses slowly, so industries with net upward analyst EPS revisions outperform over the following month, and this revision signal carries information beyond price momentum. The cross-sectional unit is the FF49 industry, the horizon is one month (signal at the end of month t , return in $t + 1$), and the test is an IC-blended, risk-targeted long-short portfolio. **Falsification:** if the revision signal or the blended strategy has no FF5+UMD alpha with $t \geq 2$ net of costs, particularly after 2010, we report “do not implement.” Windows with fewer than 36 months are reported but labelled *low power* and cannot pass or fail the test.

1.3 Data

- **Industry returns:** Ken French 49 value-weighted industry portfolios, monthly, July 1926 to August 2026 (August 2026 vintage built on the 202608 CRSP file). Returns are decimals; -99.99 codes are treated as missing.
- **Analyst revisions:** I/B/E/S summary statistics (annual EPS, current fiscal year, US firms), January 1985 to August 2026; 2.24 million firm-month snapshots. Linked to CRSP CIZ monthly data (common stock on NYSE/AMEX/Nasdaq, January 1984 to December 2025) through the WRDS I/B/E/S–CRSP link table (score ≤ 2), using the snapshot dated on or before each month-end. Firms are mapped to FF49 industries by their historical SIC code at month t , and weighted by month- t market capitalization.
- **Factors:** FF5 (2×3), UMD and short-term reversal, Ken French library, same vintage.

Two data limits matter. First, the link table has no valid links in 2026, so REV is missing from January 2026; we leave it missing rather than fill it, and report how many months each signal covers. This makes the REV-based strategies’ “recent 18 months” only 10 return months. Second, industry coverage is good but not complete: REV covers on average 93% of industries per month, and 1,037 of 24,500 industry-month cells have fewer than five linked firms (those cells are set to missing). Survivorship is handled by using CRSP’s full history (including delisted firms, with CIZ total returns) and Ken French’s historical industry portfolios. I/B/E/S summary statistics are not revised after the snapshot date, so the signal is point-in-time.

1.4 Implementation

Signals. *MOM*: compounded industry return over months $t - 11$ to $t - 1$, skipping month t . *REV*: for each firm with at least three estimates, $(\#up - \#down) / \#estimates$; industry REV is the cap-weighted mean, requiring five covered firms. *REV_alt* (robustness): three-month change in the consensus scaled by price. Each month every signal is z-scored across available industries and winsorized at ± 3 . *REV orthogonal*: the residual of a monthly cross-sectional regression of REV on MOM, which is the direct test of the thesis.

IC and blend. The monthly IC is the Spearman correlation between the signal at t and industry returns at $t + 1$; t -statistics use Newey–West (NW) standard errors with 6 lags. Blend weights follow HW2, $w \propto R^{-1}\text{IC}$, where R is the MOM–REV correlation and IC the vector of mean ICs, both estimated on an expanding window that uses only months before t (minimum 60 months), so the first blend is January 1995.

Risk model. An EWMA covariance of the 49 industries with a 30-month half-life (the HW1 choice), at least 60 months of history, estimated each month with data through t . It is shrunk 50% toward its diagonal; without shrinkage the optimizer reached 5% ex-ante risk but realized about 10% with roughly four times gross leverage. ω_n is each industry’s volatility residual to the equal-weighted industry average.

Alphas and sizing. Grinold–Kahn alphas $\alpha_n = \text{IC} \cdot \omega_n \cdot z_n$, demeaned. Holdings solve $\max_h h' \alpha - \lambda h' \Sigma h$ subject to dollar neutrality ($\sum h_n = 0$) and $|h_n| \leq 10\%$ of gross, i.e. $h = \Sigma^{-1} \alpha / (2\lambda)$ before the cap. Each month λ is set so that ex-ante active risk $\sqrt{12 h' \Sigma h}$ equals 5% a year. For the blend the median λ is 2.32 (10th–90th percentile 1.93–3.23); for MOM 3.78; for REV 1.21. Because λ rescales holdings to the target, only the sign of the strategy’s expanding IC matters, not its size. We also run the HW2 diagonal rule $h_n = \alpha_n / (2\lambda \omega_n^2)$ as a check (results in the pipeline tables; the conclusions are the same).

Timing, turnover and costs. Holdings formed at the end of month t earn $h(t)'r(t + 1)$. Turnover is $\sum_n |h_n(t) - \tilde{h}_n(t - 1)|$, where $\tilde{h}$ are last month’s weights after drifting with returns. Net returns subtract turnover times a one-way cost of 10, 20 (base case) or 30 bp, charged on the next month’s return. We run four strategies: MOM, REV, REV orthogonal and the blend. Unit tests check that signals at t align with returns at $t + 1$ and that changing future data cannot change past signals, weights or holdings.

AI prompts. We used Claude Code (Anthropic) as the coding and research assistant, and a second AI chat tool for one methodology question. The most important “prompt” was the project plan we gave the assistant: it stated the role, the thesis and rejection criterion, the data filters, the timing rules (“a signal formed at the end of month t may only use information available on or before the end of month t ”), the do-not rules, and the acceptance checks for each phase. Section 2.6 evaluates the three interactions that mattered most, and Appendix A summarizes them; the verbatim prompts and full evaluations are in 03_ai_appendix.

2 What Did We Learn?

2.1 Signal efficacy: momentum predicts, revisions do not add

Table 1: Monthly Spearman IC (signal at month-end t , industry return in $t + 1$), with Newey–West (6 lags) t -statistics. The 1985–2025 column uses only months where REV exists (492), so all signals are compared on the same months. Recent 18m = March 2025 to August 2026; REV signals cover only 10 evaluable months.

Signal	IC	NW t	IC	NW t	IC	Months
	1985–2025		Post-2010		Recent 18m (low power)	
MOM	0.047	3.96	0.035	2.22	0.015	17
REV	0.010	1.06	−0.000	−0.01	−0.010	10
REV orthogonal to MOM	−0.000	−0.05	−0.007	−0.50	0.022	10
Blend (MOM + REV)	0.040	3.17	0.036	2.49	0.018	10

Table 1 gives the central result. MOM has a mean IC of 0.047 ($t = 3.96$) over 1985–2025 and 0.035 after 2010 ($t = 2.22$). REV is positive but insignificant over the full sample (0.010, $t = 1.06$) and zero after 2010. The part of REV that is not explained by momentum has an IC of essentially zero in both windows (-0.000 , $t = -0.05$; -0.007 , $t = -0.50$). REV and MOM are positively related: their average cross-sectional correlation is 0.31, so most of what REV knows, momentum already knows.

The expanding blend weights tell the same story (Figure 1b). The weight on REV starts at 0.34 in 1990 and falls below zero by 2010 (-0.05) and to -0.13 by 2026, so the blended signal is almost pure momentum for most of the trading period. A descriptive split (chosen after seeing the weights, so not part of the test) suggests why: REV had a significant IC in 1985–1994 (0.045, $t = 2.35$) and none since 1995 (Appendix Table 6). Even in the early decade its orthogonal part was not significant (0.020, $t = 1.13$). This is consistent with revision information being priced quickly once I/B/E/S data became widely used, but we cannot test that explanation here.

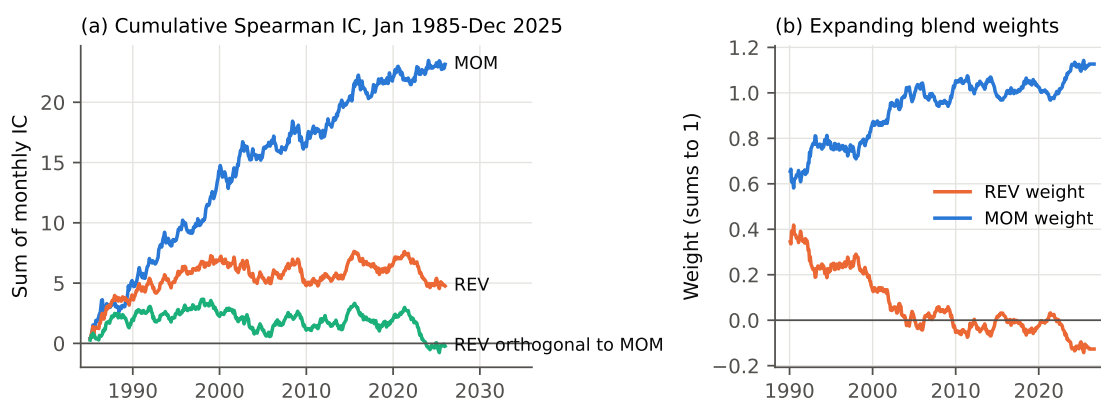


Figure 1: (a) Cumulative sum of monthly Spearman ICs over the 492 REV months, January 1985 to December 2025. A rising line means positive average IC. (b) Expanding IC-based blend weights (HW2 rule), January 1990 to August 2026; weights after 2025 are frozen because REV is missing.

2.2 Portfolio performance, risk and costs

Table 2: Mean-variance books, formation months January 1995 to December 2025 (372 months; the only window in which all four strategies trade). Returns and volatility annualized. Turnover is one-way, per month, as a fraction of capital. Break-even = one-way cost at which the mean net return is zero.

Strategy	Gross ret. (%)	Vol (%)	Gross SR	Turnover (%/mo)	Net Sharpe at			Max DD (%)	Break-even (bp)
					10 bp	20 bp	30 bp		
MOM	3.9	6.6	0.59	51	0.49	0.40	0.31	−20	63
REV	0.2	5.0	0.04	174	−0.38	−0.80	−1.21	−72	1
REV orthogonal to MOM	−0.3	4.4	−0.08	182	−0.57	−1.07	−1.56	−78	−2
Blend	3.2	6.6	0.48	58	0.38	0.27	0.17	−23	46

Performance. MOM earns 3.9% a year gross at 6.6% volatility (Sharpe 0.59) and the blend 3.2% (Sharpe 0.48); REV and REV orthogonal earn nothing gross (Table 2; cumulative returns in Appendix Figure 2). Adding REV to momentum makes the portfolio slightly worse, not better.

Turnover and costs. Industry revision ranks change a lot from month to month, so REV books trade about 174% of capital a month (182% for REV orthogonal), against 51% for MOM and 58% for the blend. REV’s break-even cost is about 1 bp, so at 20 bp it loses 4.0% a year (net Sharpe -0.80) with

a 72% drawdown. Costs therefore destroy whatever small edge revisions might have. The blend survives costs in raw terms (net Sharpe 0.27 at 20 bp), but as the next section shows, that is because it is momentum.

Risk. Ex-ante active risk equals the 5% target every month by construction. Realized volatility is 6.6% for the blend (1.32 times target) and 6.2% for MOM, mainly because of the 2009 momentum crash; the REV books realize 4.3–4.8%. Gross exposure averages 1.2 (MOM, blend) to 1.5 (REV books), and the blend’s worst net drawdown is 23% (October 2012). Full risk statistics are in Appendix Table 7.

2.3 Factor attribution: the alpha is momentum beta

Table 3: Net returns (20 bp) regressed on FF5, FF5+UMD (primary) and FF5+UMD+short-term reversal (STREV). Annual alpha in %, Newey–West (6 lags) t in parentheses. Post-2010 has 192 return months for REV-based books (REV ends December 2025) and 199 for MOM.

Strategy	Window	n	FF5 α (t)	FF5+UMD α (t)	UMD β (t)	R^2 FF5 / +UMD	+STREV α (t)
MOM	1995–2025	372	3.3 (2.84)	1.2 (1.55)	0.29 (16.9)	0.11 / 0.56	1.2 (1.53)
MOM	Post-2010	199	2.2 (1.68)	0.7 (0.64)	0.27 (11.1)	0.16 / 0.47	0.8 (0.69)
REV	1995–2025	372	−3.5 (−4.49)	−4.5 (−5.83)	0.13 (7.6)	0.08 / 0.25	−4.5 (−5.78)
REV	Post-2010	192	−3.0 (−2.72)	−3.5 (−3.02)	0.09 (3.2)	0.08 / 0.13	−3.6 (−3.12)
REV orthogonal to MOM	1995–2025	372	−4.2 (−5.11)	−4.6 (−5.50)	0.07 (4.3)	0.04 / 0.09	−4.6 (−5.49)
REV orthogonal to MOM	Post-2010	192	−3.3 (−2.80)	−3.7 (−2.94)	0.06 (2.7)	0.03 / 0.06	−3.7 (−2.99)
Blend	1995–2025	372	2.4 (2.21)	0.2 (0.36)	0.30 (20.4)	0.10 / 0.59	0.2 (0.35)
Blend	Post-2010	192	1.6 (1.28)	−0.0 (−0.02)	0.29 (13.8)	0.13 / 0.46	0.0 (0.03)

This is the decisive test. Against FF5 alone, both MOM and the blend look like alpha: 3.3% a year ($t = 2.84$) for MOM and 2.4% ($t = 2.21$) for the blend over 1995–2025 (Table 3; Appendix Figure 3). Adding UMD removes most of it. The blend’s alpha falls to 0.2% ($t = 0.36$) with a UMD loading of 0.30 ($t = 20.4$), and R^2 rises from 0.10 to 0.59. After 2010 the blend’s FF5+UMD alpha is −0.0% ($t = −0.02$). Adding short-term reversal changes nothing. REV and REV orthogonal have significantly *negative* net alphas (−4.5% a year, $t \approx −5.5$ to $−5.8$), and both load positively on UMD (0.13 and 0.07): even after orthogonalizing against our industry momentum, the revision book still tilts toward stock-level momentum. A strategy that looks good against FF5 and loses its alpha once UMD is included is repackaging momentum; that is what the blend does.

Gross alpha and the cost line. Before costs the blend has a 6-factor alpha of 1.6% a year ($t = 2.44$, 1995–2025), less than MOM alone (2.4%, $t = 3.18$), so revisions dilute rather than add (Appendix Table 8). The blend’s alpha is fully consumed at 23 bp one-way, is not significant at 10 bp ($t = 1.40$), and is not significant after 2010 even gross ($t = 1.26$). A side result we report but do not claim: plain industry momentum keeps some alpha beyond UMD at 10 bp (1.8%, $t = 2.36$, 1995–2025), but not at our 20 bp base case ($t = 1.55$) or after 2010 ($t = 0.64$). Our Climate work found the same pattern for an optimized momentum book, so we see it as a paper-trading candidate, not a result.

2.4 Time patterns: full sample, post-2010 and the last 18 months

The blend’s net Sharpe is 0.27 over 1995–2025 and 0.32 after 2010, and its FF5+UMD alpha is zero in both (Table 4). In the most recent window the blend shows a 6-factor alpha with $t = 3.40$, but that regression has 10 monthly observations for seven parameters and six NW lags; the t -statistic carries no information and we report it only because the brief requires the window. MOM’s recent window (17 months) has an alpha of −0.5% ($t = −0.12$).

Table 4: Horizon analysis, net of 20 bp. Windows by formation month; “Full” starts when each book first trades (MOM 1974, REV 1990, blend 1995). Regressions with fewer than 36 months are LOW POWER and not used for the verdict. REV rows are in tables/T6_horizons.csv.

Strategy	Window	Return months	n	Net ret. (%)	Net SR	FF5+UMD α (t)	
MOM	Full	1974-07–2026-08	626	2.1	0.35	0.4 (0.70)	
MOM	1995–2025	1995-02–2026-01	372	2.7	0.40	1.2 (1.55)	
MOM	Post-2010	2010-02–2026-08	199	2.5	0.44	0.7 (0.64)	
MOM	Recent 18m	2025-04–2026-08	17	1.9	0.28	−0.5 (−0.12)	LOW POWER
Blend	Full	1995-02–2026-01	372	1.8	0.27	0.2 (0.36)	
Blend	1995–2025	1995-02–2026-01	372	1.8	0.27	0.2 (0.36)	
Blend	Post-2010	2010-02–2026-01	192	1.7	0.32	−0.0 (−0.02)	
Blend	Recent 18m	2025-04–2026-01	10	2.8	0.52	3.3 (3.40)	LOW POWER

Performance is regime-dependent. In 2009 the blend lost 13.3% and MOM 11.8% while UMD lost 52.8%; April 2009 alone cost the blend 12.1%. REV did not hedge the crash: it lost 18.3% that year. In 2020 the blend gained 12.3% and MOM 14.3% (Appendix Table 9). The rolling 36-month FF5+UMD alpha of the blend averages −0.1% a year and ranges from −6.4% to +5.3% over 337 windows; 6% of windows have $t \geq 2$ and 5% have $t \leq -2$, about what noise produces in overlapping windows (Appendix Figure 4). There is no persistent alpha regime.

2.5 Robustness

Table 5: One-at-a-time robustness grid for the blended strategy. Each row reruns signals, IC blend, holdings and backtest; all rows are scored on formation months January 1995 to December 2025, net of 20 bp unless the row changes costs. (*) Run by a teammate on a different cleaned CRSP panel; not strictly comparable and not reproducible without WRDS stock-level files.

Change from base	Gross SR	Net SR	Net SR post-2010	Turnover (%/mo)	FF5+UMD α (%)	t
Base case	0.48	0.27	0.32	58	0.2	0.36
Costs 10 bp	0.48	0.38	0.45	58	0.9	1.40
Costs 30 bp	0.48	0.17	0.20	58	−0.5	−0.67
Momentum lookback 6m	0.15	−0.40	−0.15	130	−3.1	−3.89
Momentum lookback 9m	0.32	0.04	0.11	75	−1.3	−1.83
REV = consensus change	0.38	0.15	0.20	62	−0.6	−0.82
Cov. half-life 18m	0.50	0.28	0.34	59	0.3	0.38
Cov. half-life 60m	0.45	0.26	0.31	57	0.1	0.18
Risk target 3%	0.48	0.27	0.32	35	0.1	0.36
Risk target 8%	0.48	0.27	0.32	93	0.4	0.36
Dollar + beta neutral	0.56	0.33	0.34	60	0.5	0.74
Min analysts 1 (*)	0.54	0.34	0.45	56	0.8	1.07
Min analysts 5 (*)	0.41	0.19	0.26	61	−0.4	−0.64
30 industries (*)	0.54	0.34	0.49	52	0.7	0.87

The question is whether any reasonable design choice changes the conclusion (Table 5). None does. The highest FF5+UMD t in the grid is 1.40 (10 bp costs) and the highest net Sharpe 0.38. The covariance half-life (18 to 60 months) and the risk target (3% to 8%) barely matter: the risk target only scales positions and turnover. Shorter momentum lookbacks trade more and lose the edge (6-month lookback: alpha −3.1%, $t = -3.89$). Replacing the revision measure with the consensus

change makes the blend worse. A dollar- and beta-neutral book has $t = 0.74$. A full 3×3 grid of momentum lookback and covariance half-life (Appendix Figure 5) gives net Sharpe ratios between -0.42 and 0.28 .

2.6 Critical evaluation of the AI assistant

What worked. Given a precise plan, the assistant wrote a correct, tested pipeline quickly: point-in-time joins, expanding-window blends, a constrained optimizer, drift-adjusted turnover and Newey–West attribution, with unit tests for every timing rule. It also caught problems we had not asked about. In its review of our Phase 2–3 code it found that momentum used months $t - 12$ to $t - 2$ instead of the planned $t - 11$ to $t - 1$, and that CRSP characteristics were being carried across gaps; at the Phase 4 checkpoint it flagged early that the orthogonal REV IC was zero.

What was wrong. The errors were mostly about judgment and environment, not syntax. (1) When we said “ignore 2026,” it added a December 2025 sample cutoff, which silently moved the required “last 18 months” to July 2024–December 2025, exactly the kind of window-shifting the brief warns against. We reversed it and report thin coverage instead. (2) Its rejection-criterion flag marked the blend’s 10-month recent window as a pass ($t = 3.40$); it had no guard for tiny samples, so we added the 36-month low-power rule. Its first draft summary then claimed no window reached $t \geq 2$, which was also false. (3) It tested only in its own Python 3.11 / pandas 3 sandbox, so its type hints crashed on a teammate’s Python 3.9, and a row-alignment bug in the attribution code crashed under pandas 2.2. (4) When the risk model missed its target, it diagnosed the problem but left the shrinkage choice to us, which was right.

What we changed in how we worked. Our follow-up prompts were often one word (“ok” to start the next phase), which delegated decisions to the plan and to the model. That worked only because the plan fixed the thesis, the windows and the rejection rule in advance. Where the plan was silent (the 2026 gap, small samples) the model filled the gap with a plausible but wrong default. The lesson for an investment research assistant: it is an excellent implementer and a useful second reader, but it optimizes for a clean-looking result unless the rules forbid it, and it does not check its assumptions against the user’s environment.

2.7 Decision

Do not implement. The pre-registered test fails for every strategy in the 1995–2025 and post-2010 windows and for every robustness row. The economic reason is clear in the data: industry revisions are mostly a lagged reflection of the same news that drives industry momentum (correlation 0.31, zero orthogonal IC), the small independent part they had in the late 1980s disappeared, and their rank volatility makes them expensive to trade. For the revision idea to work, one would need a signal that changes more slowly than monthly revision counts (for example, a longer revision window or firm-level drift) and evidence that it predicts returns after UMD, net of realistic costs.

Limitations. We test industry portfolios, not firms, so firm-level revision drift could still exist and be diversified away by aggregation. REV ends in December 2025, so the recent window is thin. The 50% covariance shrinkage was chosen after seeing that the unshrunk model missed its risk target (a risk decision, not a return-based one). Costs are linear with no market impact, and three robustness rows could not be rerun in the final environment and are flagged.

A AI interactions (summary)

Full prompts (verbatim), output summaries and evaluations are in 03_ai_appendix/AI_Interactions.pdf. The table summarizes the three required uses.

Use	Prompt (excerpt)	What was wrong or incomplete	What we changed / decided
1. Research design and data review (Claude Code; a second AI chat for one question)	Plan file with thesis, rejection rule and timing rules; “how is this looking so far; any improvements needed?”; “currently ignoring 2026”	Added a December 2025 cutoff that shifted the required recent window; earlier code had a $t - 12$ to $t - 2$ momentum window	Window kept at March 2025–August 2026 with coverage reported; momentum fixed to $t - 11$ to $t - 1$; future-data invariance test added
2. Coding and implementation (risk model, optimizer, backtest)	“yes do steps 1-2 then start phase 5”; “ok go with covariance shrinkage and go with phase 6”	Realized risk $2\times$ target with the unshrunk covariance; Python 3.10-only type hints; tests only in its own environment	Team chose 50% shrinkage; code made Python 3.9-compatible and retested on a teammate’s machine
3. Robustness and attribution design	“ok” (start Phase 7 and Phase 8 as planned)	Low-power window flagged as a pass; draft summary claimed no $t \geq 2$; alignment bug under pandas 2.2	36-month low-power rule; summary corrected; alignment fixed with a regression test

B Additional tables and figures

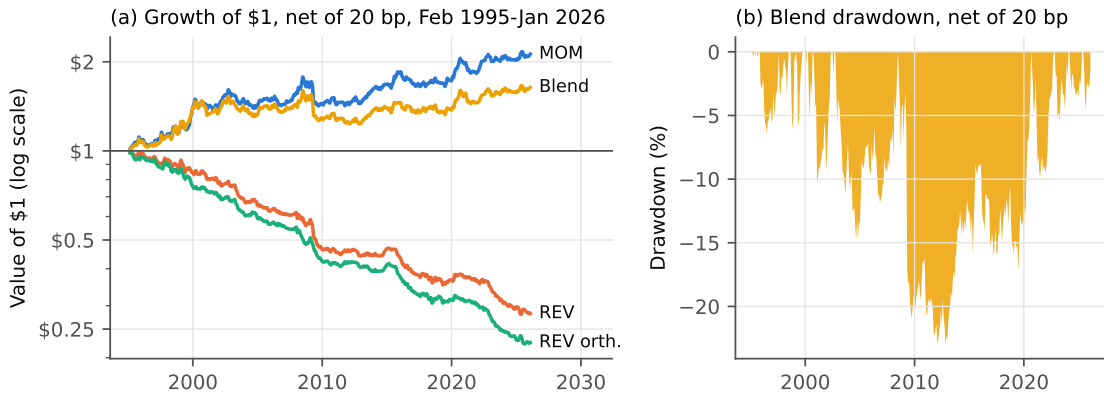


Figure 2: (a) Growth of \$1 invested in each mean-variance book, net of 20 bp one-way costs, return months February 1995 to January 2026, log scale. (b) Drawdown of the blended book, net of 20 bp.

Table 6: Descriptive IC split (Spearman, NW t in parentheses), REV months only. Chosen after seeing the blend weights; not part of the test.

Signal	1985-1994	1995-2009	2010-2025
MOM	0.071 (3.06)	0.043 (1.90)	0.036 (2.23)
REV	0.045 (2.35)	-0.003 (-0.21)	-0.000 (-0.01)
REV orthogonal to MOM	0.020 (1.13)	-0.007 (-0.53)	-0.007 (-0.50)

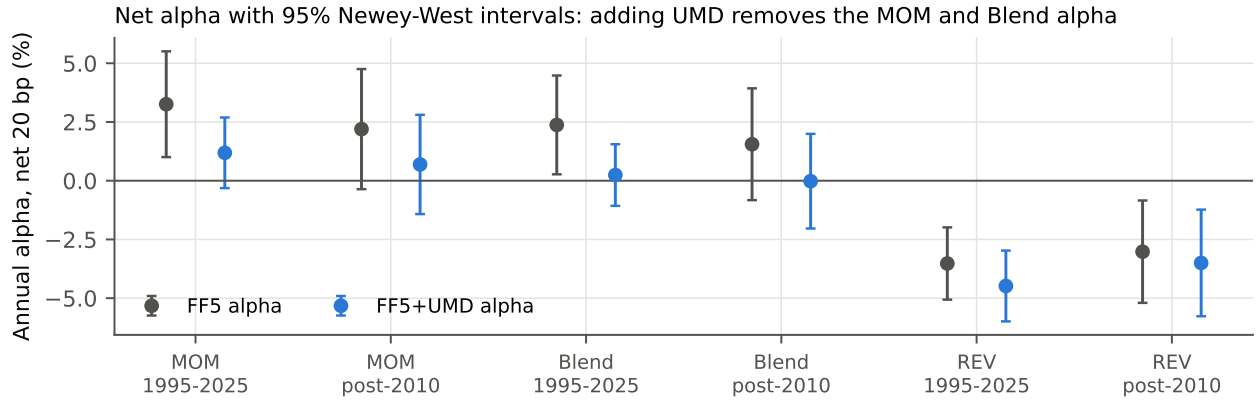


Figure 3: Annual net alpha (20 bp) with 95% Newey–West intervals against FF5 (gray) and FF5+UMD (blue), 1995–2025 and post-2010 windows.

Table 7: Risk and sizing, mean-variance books, each from its first month to its last. Ex-ante risk is the mean of the monthly ex-ante active risk; realized is the annualized volatility of monthly gross returns.

Strategy	First month	Ex-ante (%)	Realized (%)	Ratio	Median λ (p10–p90)	Gross
MOM	1974-06	5.0	6.2	1.24	3.78 (3.28–4.30)	1.20
REV	1990-01	5.0	4.8	0.96	1.21 (0.72–2.50)	1.50
REV orthogonal to MOM	1990-01	5.0	4.3	0.86	0.33 (0.14–1.12)	1.53
Blend	1995-01	5.0	6.6	1.32	2.32 (1.93–3.23)	1.22

Table 8: FF5+UMD alpha (% a year, NW t) by cost level, and the one-way cost at which the gross 6-factor alpha is fully consumed.

Strategy	Window	Gross	Net 10 bp	Net 20 bp	Net 30 bp	Zero- α cost (bp)
MOM	1995–2025	2.4 (3.18)	1.8 (2.36)	1.2 (1.55)	0.6 (0.74)	39
Blend	1995–2025	1.6 (2.44)	0.9 (1.40)	0.2 (0.36)	–0.5 (–0.67)	23
REV	1995–2025	–0.3 (–0.39)	–2.4 (–3.10)	–4.5 (–5.83)	–6.6 (–8.54)	–1
REV orthogonal to MOM	1995–2025	–0.3 (–0.30)	–2.4 (–2.90)	–4.6 (–5.50)	–6.8 (–8.11)	–1
MOM	Post-2010	1.9 (1.76)	1.3 (1.20)	0.7 (0.64)	0.1 (0.09)	32
Blend	Post-2010	1.3 (1.26)	0.6 (0.62)	–0.0 (–0.02)	–0.7 (–0.65)	20
REV	Post-2010	0.8 (0.68)	–1.4 (–1.18)	–3.5 (–3.02)	–5.6 (–4.84)	4
REV orthogonal to MOM	Post-2010	0.8 (0.67)	–1.4 (–1.14)	–3.7 (–2.94)	–5.9 (–4.74)	4

Table 9: Stress years: compounded net returns (20 bp) of each book and the UMD factor, in %.

Year	Blend	MOM	REV	REV orth.	UMD factor
2009	–13.3	–11.8	–18.3	–14.7	–52.8
2020	12.3	14.3	2.0	–1.3	–0.9

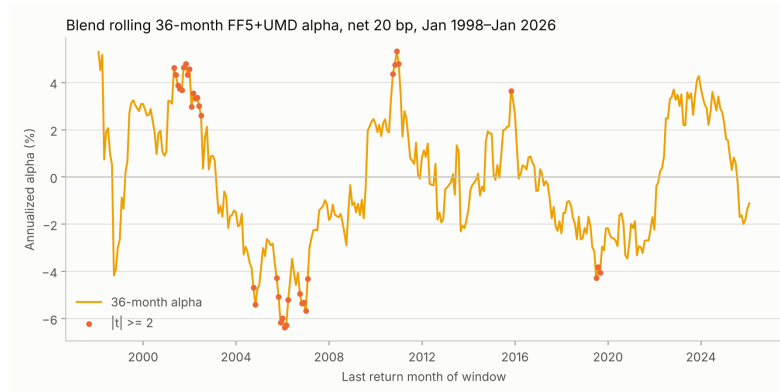


Figure 4: Rolling 36-month FF5+UMD alpha of the blended book, net of 20 bp, windows ending January 1998 to January 2026 (pipeline output).

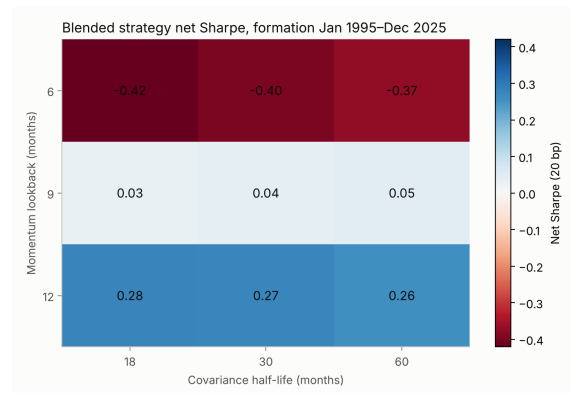


Figure 5: Net Sharpe (20 bp) of the blended book over momentum lookback and covariance half-life, formation months 1995–2025 (pipeline output).

C Reproducibility and code

MFE230GA_Final_Analysis.ipynb reruns Phases 4–7 from the committed industry signal panel and rebuilds every table and figure in this report in about two minutes; the robustness grid takes about seven minutes more (RUN_ROBUSTNESS=True). Phases 1–3 need WRDS access. All parameters are in config.py; 67 unit tests pass and one git-only test is skipped (python -m pytest tests). Every number in the text is written to tables/report_numbers.json and traced in CLAIM_AUDIT.md. The core sizing step:

```
# src/portfolio.py, build_holdings() (abridged): one month, one strategy
cov_ann = risk.shrink_covariance(covs[month], shrinkage) * 12 # 50% toward diagonal, annualized
omega = risk.residual_volatility(covs[month])[available] # vol residual to EW industry average
alpha = grinold_kahn_alpha(z[available], ic, omega) # IC * omega * z, demeaned
unit, converged = capped_mean_variance(alpha, sub_cov) # lambda = 1; sum(h)=0, |h_n| <= 10% of gross
h, lam, exante = calibrate_to_target(unit, sub_cov, 0.05) # lam = sqrt(unit' S unit)/0.05; h = unit/lam
```

Audit changes made for this submission. All pipeline outputs were regenerated from the committed signal panel and matched the committed tables to within 10^{-8} . We added the descriptive IC split, gross and cost-level 6-factor alphas, break-even costs and an independent re-estimate of the key regressions. Three robustness rows produced on a different CRSP panel are now flagged rather than mixed with the base rows. One unit test that requires a git repository now skips outside one.